Speech Processing

1. Consider a discrete-time speech signal $s(n)$ corrupted by additive noise $\nu(n)$. The noisy speech signal can be written in the time domain as $y(n) = s(n) + \nu(n)$.

   a) Derive in full detail the Weiner filter that could be used to reduce the noise and state under what conditions the Weiner filter is optimal. [4]

Solution:
Weiner filter provides the min mean square solution to the optimal filtering problem.
Solution must first define the error signal vector and the filter matrix $\mathbf{H}$

$$\begin{aligned}\mathbf{e}(n) &= \mathbf{z}(n) - \mathbf{s}(n) \\ &= \mathbf{H}(n)[\mathbf{s}(n) + \mathbf{v}(n)] - \mathbf{s}(n)\end{aligned}$$

and then set up the MMSE cost function

$$\begin{aligned}J(\mathbf{H}) &= \mathrm{tr}\left\{E[\mathbf{e}(n)\mathbf{e}^T(n)]\right\} \\ &= \mathrm{tr}(\mathbf{R}_s) + \mathrm{tr}(\mathbf{H}\mathbf{R}_y\mathbf{H}^T) - 2\mathrm{tr}(\mathbf{H}\mathbf{R}_s)\end{aligned}$$

where

$$\mathbf{R}_s = E[\mathbf{s}(n)\mathbf{s}^T(n)] \text{ and } \mathbf{R}_y = E[\mathbf{y}(n)\mathbf{y}^T(n)].$$

The next step is to optimize by setting the partial derivatives to zero so that

$$\frac{\partial J}{\partial \mathbf{H}} = 0$$

leading to

$$\mathbf{H}_w^T = \mathbf{R}_y^{-1}\mathbf{R}_s = \mathbf{I} - \mathbf{R}_y^{-1}\mathbf{R}_v$$

   b) In a wideband telecommunication system, a noisy speech signal $y(n)$ is obtained from a single microphone and in the frequency domain can be written as

$$Y(l,k) = S(l,k) + V(l,k)$$

   for time-frame index l and frequency index k.

   i) Explain what is meant by spectral variance, $\phi_y(l,k)$, of the noisy speech signal. [2]

   ii) Write down the mathematical definition for $\phi_y(l,k)$. [4]

   iii) Describe in detail a method based on Voice Activity Detection, and a method based on Minimum Statistics, to compute an estimate of the spectral variance of the noise, $\hat{\phi}_v(l,k)$. Include mathematical expressions for the estimators in your description where relevant. [4]

iv) Compare and contrast the advantages and disadvantages of these alternative methods. [2]

Solution:

(i)

For any signal $x(n)$ with time-frequency domain representation $X(l,k)$, the spectral variance is a measure of the signal power at time-frame l and frequency k.

(ii)

$$\phi_y(l,k) = E\left[|Y(l,k)|^2\right]$$

(iii)

Voice activity detection (VAD): determine the segments of the signal during which only noise is present. Then update the noise model according to:

$$\hat{\phi}_v(l,k) = \begin{cases} \hat{\phi}_v(l-1,k), & \text{if speech is active} \\ \alpha\hat{\phi}_v(l-1,k) + (1-\alpha)|Y(l,k)|^2, & \text{if speech is not active.} \end{cases}$$

The term α has to be chosen according to the desired level of smoothing and can be time and frequency varying.

Minimum statistics approach: This technique is based on the assumption that during a speech pause, or within brief periods between words and even syllables, the speech energy is close to zero. As a result, a short-term power spectrum estimate of the noisy signal, even during speech activity, decays frequently due to the noise power. Thus, by tracking the temporal spectral minimum without distinguishing between speech presence and speech absence, the noise power in a specific frequency band can be estimated.

$$\hat{\phi}_y(l,k) = \alpha\hat{\phi}_y(l-1,k) + (1-\alpha)|Y(l,k)|^2$$
$$\hat{\phi}_v(l,k) = \min\left\{\hat{\phi}_y(l,k),\ \hat{\phi}_y(l-1,k),\ \ldots,\ \hat{\phi}_y(l-D+1,k)\right\}$$

(iv)

The VAD method compared to the MS method is able to estimate the noise simply and accuratly but only if the VAD is itself accurate and reliable - this is rarely the case especially in the presence of noise. Unlike the MS method, the VAD method cannot update its noise estimate except during pauses and therefore cannot normally track time-varying noise well. In such cases, the performance of the MS method is much better.

c) Consider a noise reduction algorithm applied to the noisy speech signal. The device uses the method of amplitude spectral subtraction.

Show that the output signal of the noise reduction algorithm at time-frame l and frequency index k can be written

$$Z(l,k) = H(l,k)Y(l,k)$$

and write down an expression for $H(l,k)$ in terms of $\hat{\phi}_v(l,k)$. [4]

Solution:
For amplitude spectral subtraction, we have

$$\begin{aligned} Z(l,k) &= Y(l,k) - \sqrt{\hat{\phi}_v(l,k)} \\ &= Y(l,k)\left(1 - \frac{\sqrt{\hat{\phi}_v(l,k)}}{Y(l,k)}\right) \\ &= Y(l,k)H(l,k) \end{aligned}$$

2. a) In evaluating a speech recognition system, state the 3 types of word-errors that are considered when computing the word error rate. Explain each type of error, and give the formula for the word error rate. [4]

Solution:
substitution: a wrong word replaces the correct word
insertion: an additional word is erroneously inserted into the text that does not appear in the signal
deletion: a word that does appear in the signal is erroneously omitted from the text

$$\text{Word Error Rate (WER)} = \frac{\#\text{substitutions} + \#\text{insertions} + \#\text{deletions}}{\#\text{words}}\%$$

b) Consider a speech recognition system based on Deep Neural Networks (DNN). Data augmentation is a common strategy to increase the quantity of training data, and improve the robustness of the trained DNN models.

i) State 3 ways in which speech signals can be modified to achieve augmentation of the training data. [3]

ii) Explain why such augmentation can improve the performance of a speech recognition system. [2]

Solution:
Any 3 of the following:

- change the gain of the signal
- change the speaking rate
- add noise
- add reverb by convolving with a range of room impulse responses
- pass the speech signal through a speech CODEC, or various CODECs, a different data rates
- generate new speech using text-to-speech

The performance improvement comes from improving the chances that the trained model matches well with the test data at inference.

c) Consider a speech recognition system based on Hidden Markov models (HMM).

i) For an arbitrary state, s, in the HMM, let the probability of a transition from state s to the next state be denoted p. Taking into account only this transition probability p, derive an expression in terms of p for the number of frames aligned to state s. Show all the details of your derivation. [4]

Solution:
(i)

$$pr(D=n) = p(1-p)^{n-1} \Longrightarrow E(D) = \sum_{n=1}^{\infty} np(1-p)^{n-1} = \frac{1}{p}$$

since

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

and by differentiating both sides

$$\sum_{n=0}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

ii) n short speech utterance is represented by feature vectors $\mathbf{x}_1$,$\mathbf{x}_2$,$\mathbf{x}_3$,$\mathbf{x}_4$,$\mathbf{x}_5$. These feature vectors are to be compared with a 3-state HMM for which the output probability densities of the feature vectors for each state are shown in Table 1. The state diagram of the HMM is shown in Figure 2.1, in which the labels on the arrows indicate the state transition probabilities.

Construct the lattice diagram of the model and draw on the lattice all the feasible paths through the lattice. [2]

Determine the maximum probability alignment of the frames to the states of the HMM and the value of B(3,5) corresponding to this alignment. [5]

	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
State 1	0.5	0.2	0.6	0.4	0.5
State 2	0.5	0.7	0.3	0.1	0.5
State 3	0.5	0.5	0.4	0.6	0.6

Table 1

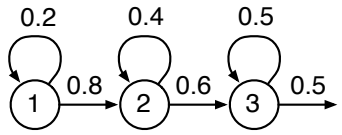
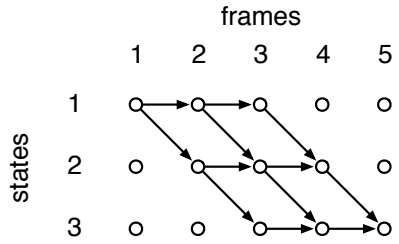


Figure 2.1

Solution:
(ii)



$B(1,1) = \mathbf{0.5}$
$B(1,2) = 0.5 \times 0.2 \times 0.2 = 0.02$
$B(2,2) = 0.5 \times 0.8 \times 0.7 = \mathbf{0.28}$
$B(1,3) = 0.02 \times 0.2 \times 0.6 = 0.0024$
$B(2,3) = \max(0.28 \times 0.4 \times 0.3, 0.02 \times 0.8 \times 0.3) = 0.0336$
$B(3,3) = 0.28 \times 0.6 \times 0.4 = \mathbf{0.0672}$
$B(2,4) = 0.0336 \times 0.4 \times 0.1 = 0.0013$
$B(3,4) = \max(0.0336 \times 0.6 \times 0.6, 0.0672 \times 0.5 \times 0.6) = \mathbf{0.0202}$
$B(3,5) = \max(0.0013 \times 0.6 \times 0.6, 0.0202 \times 0.5 \times 0.6) = \mathbf{0.0061}$

and the alignment path for the five frames is 1,2,3,3,3.

3. a) A linear predictor for the speech signal $s(n)$ can be written

$$s(n) = Gu'(n) + \sum_{j=1}^{p} a_j s(n-j).$$

i) Describe how Linear Predictive Coding (LPC) is used in speech signal processing applications. Include any relevant diagrams and mathematical formulae. [5]

ii) Explain what the terms

$$u'(n),\ G,\ \text{and}\ \ a_j$$

represent in this expression. [3]

Solution:
(i)
LPC is used as a method of spectral estimation from which characteristic features of speech can be obtained. The LPC coefficients are the denominator coefficients of an all pole (AR) transfer function. A close link can be derived between the AR system and the vocal tract transfer function. The LPC coefficients may be used in speech coding (often after conversion to Line Spectral Frequencies) and speech recognition (often after conversion to cepstral coefficients.)
(ii)
$u'(n)$ represents the derivative of glottal flow.
G represents a gain factor.
a_j represent the predictor coefficients, or the coefficients of the denominator polynomial of the vocal tract transfer function.

b) For a linear predictor, and stating any assumptions used, formulate an expression for the prediction error and hence show that the prediction error is minimum when the following expression is satisfied: [3]

$$\sum_{j-1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1,\ \ldots,\ p.$$

Explain the meaning of $\{F\}$. [1]

Solution:
Under the assumption that the second term dominates, the prediction error is given by

$$e(n) = s(n) - \sum_{j=1}^{p} a_j s(n-j).$$

The sum squared prediction error is $Q_E = \sum_{n\in\{F\}} e^2(n)$ which is differentiated w.r.t. each a_j giving

$$\frac{\partial Q_E}{\partial a_i} = \sum_{n\in\{F\}} \frac{\partial \left(e^2(n)\right)}{\partial a_i} = -\sum_{n\in\{F\}} 2e(n)s(n-i).$$

The optimum values of a_j are then found from

$$\sum_{n\in\{F\}} e(n)s(n-i) = 0 \quad \text{for} \quad i = 1,\ \ldots,\ p$$

$$\sum_{n\in\{F\}} \left(s(n)s(n-i) - \sum_{j=1}^{p} a_j s(n-j)s(n-i) \right) = 0 \quad \text{for} \quad i = 1,\ \ldots,\ p$$

$$\sum_{j=1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1,\ \ldots,\ p.$$

Here, $\{F\}$ denotes the set of speech samples within a frame.

c) i) Define the complex cepstral coefficients of a speech signal and briefly state their important properties for speech processing. [2]

ii) Derive a recursion to obtain the complex cepstral coefficients from the terms a_j as defined by part a) of this question. [4]

iii) Find the complex cepstral coefficients for the case when

$$[a_0,\ a_1,\ a_2,\ a_3,\ a_4] = [1,\ -0.35,\ -0.25,\ 0.18,\ 0.4].$$

[2]

Solution:

(i)
Cepstrum is defined as inverse Fourier transform of log spectrum.

$$c_n = \frac{1}{2\pi}\int_{\omega=-\pi}^{\pi} \log\left(V(e^{j\omega})\right) e^{j\omega n} d\omega.$$

It can be computed either from roots of the prediction filter polynomial or from the coefficients of the prediction filter polynomial:

- good at discriminating between different phonemes
- fairly independent of each other
- have approximately Gaussian distributions for a particular phoneme.

(ii)

$$\begin{aligned}
&C(z) = \log(G) - \log(A(z)) \\
&A(z)zC'(z) = -zA'(z) \\
&\left(1 - \sum_{k=1}^{p} a_k z^{-k}\right)\left(z\sum_{m=0}^{\infty} -mc_m z^{-(m+1)}\right) = -z\sum_{n=1}^{p} na_n z^{-(n+1)} \\
&\left(1 - \sum_{k=1}^{p} a_k z^{-k}\right)\left(\sum_{m=1}^{\infty} mc_m z^{-m}\right) = \sum_{n=1}^{p} na_n z^{-n} \\
&\sum_{n=1}^{\infty} nc_n z^{-n} - \sum_{k=1}^{p}\sum_{m=1}^{\infty} mc_m a_k z^{-(m+k)} = \sum_{n=1}^{p} na_n z^{-n} \\
&\sum_{n=1}^{\infty} nc_n z^{-n} = \sum_{n=1}^{p} na_n z^{-n} + \sum_{k=1}^{p}\sum_{n=k+1}^{\infty} (n-k)c_{(n-k)} a_k z^{-n}.
\end{aligned}$$

Now take the coefficient of z^{-n} in the above equation noting that

$$n \geq k+1 \Rightarrow k \leq n-1$$

to give

$$\begin{aligned}
nc_n &= na_n + \sum_{k=1}^{\min(p,n-1)} c_{(n-k)} a_k \\
\Rightarrow c_n &= a_n + \frac{1}{n}\sum_{k=1}^{\min(p,n-1)} c_{(n-k)} a_k
\end{aligned}$$

(iii)

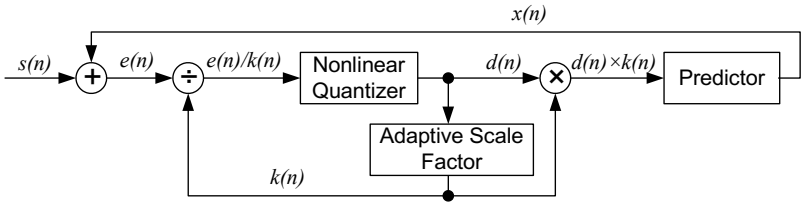
$$c = \begin{pmatrix} 1.0000 & 0.1500 & -0.2667 & -0.1088 & 0.3663 \end{pmatrix}$$

4. a) Consider a Differential Pulse Code Modulation (DPCM) speech coder.

i) Draw a fully labelled block diagram of a DPCM speech coder. [3]

ii) Describe how each of the blocks in the diagram operates to achieve a reduction in bit rate while maintaining signal quality. [3]

Solution:



The non-linear quantizer has quantization levels that are optimal for a fixed-variance Gaussian pdf. The quantization levels are closer together at low signal levels: this reduces the mean square quantization error since these levels occur with higher probability.

The adaptive scale factor normalizes the input to the quantizer so that its variance remains approximately constant at the value required by the quantizer design. The scale factor is reduced each time the quantizer output gives zero and increased each time the quantizer output gives a large value. The scale factor thus converges to a value proportional to the RMS value of $e(n)$ and the scaled quantization error will be proportional to the variance of $e(n)$.

The predictor uses previously transmitted information to predict the value of s(n). If the prediction is a good one, the signal e(n) will have a smaller variance than s(n) and the quantisation error will be correspondingly reduced. The ratio of the variances is the prediction gain.

b) Consider a uniform quantizer having quantization intervals of width w. Show that the mean square quantization error is given by

$$\frac{w^2}{12}.$$

State any assumptions used. [4]

Solution:
Assuming that the quantisation error is uniformly distributed in the range $\pm w/2$ with a pdf $p(e) = w^{-1}$, the mean square quantisation error is given by

$$\int_{-w/2}^{w/2} \frac{e^2}{w} de = \left[\frac{e^3}{3w}\right]_{-w/2}^{w/2} = \frac{w^2}{12}.$$

c) Consider a uniform quantizer with output levels at values

$$\pm 64(2k-1) \text{ for } k = 1, \ldots, 128.$$

i) Using the formula from above (question part b) calculate the mean square quantization error for this quantizer. [2]

ii) Find the mean square value of a uniformly distributed signal occupying the range of amplitude ± 11200. [2]

iii) Hence, calculate the signal-to-quantization-noise ratio in dB. [2]

Solution:
(i) Quantization interval = 128 and hence mean square error = $128^2/12 = 1365.3$.
(ii) Mean square value of the signal is $1/12 \times 22400^2 = 4.1813 \times 10^7$.
(iii) SNR $= 10\log_{10}(4.1813 \times 10^7/1365.3) = 44.86$ dB.

d) i) Non-uniform quantization schemes may be advantageous for some signals. Briefly explain why and state the relevant signal characteristics. [2]

ii) Now consider a particular non-uniform quantizer with output levels at values given by

$$\pm(2^e(2m+33)-33), \quad m = 0,1,\ldots,15 \text{ and } e = 0,1,\ldots 7.$$

Find the smallest and the largest quantization step. [2]

Solution:
(i) Matching the quantization steps to the pdf of the signal gives a lower quantization error on average, so those signal amplitudes that most commonly occur are sampled with higher precision than rarely occurring signal amplitudes.
(ii) Smallest quantization step with $e = 0$ is given by

$$[2^0((2+33)-33)] - [2^0((0+33)-33)] = 2.$$

Largest quantization step with $e = 7$ is given by

$$[2^7((2+33)-33)] - 2^7((0+33)-33)] = 256.$$